■ Apex Manufacturing Analytics

Power BI + SQL Data Warehouse

End-to-End Manufacturing Analytics & Business Intelligence Project

SQL SERVER	POWER BI	DAX	AI	PROJECT
Data Warehouse	Visualization	Analytics	Power BI	Completed

■ Project Overview

Apex Manufacturing Analytics is an end-to-end Business Intelligence project developed using **SQL Server and Microsoft Power BI**. The project transforms manufacturing, supplier, purchasing, and inventory data into actionable business insights.

- SQL Data Warehousing
- Power BI dashboards
- DAX-based KPI analysis
- AI-powered analytics
- SQL-to-Power BI validation
- Data-driven business recommendations

■ Business Problem

Manufacturing organizations generate large volumes of operational and supply-chain data. Raw data does not immediately answer which operating conditions are associated with machine failures, which failure types occur most frequently, which suppliers have fulfillment or delivery issues, which warehouses have higher reorder risk, or which operating combinations should receive preventive maintenance.

This project addresses these questions through an end-to-end analytics solution.

■ Project Architecture

```
Raw Manufacturing Data
        ↓
Staging Layer
        ↓
SQL Data Warehouse
        ↓
Fact & Dimension Data
        ↓
Analytical SQL Views
        ↓
Power BI Model
        ↓
Interactive Dashboards
        ↓
AI-Powered Analysis
        ↓
Business Recommendations
```

■■ SQL Data Warehouse

The project uses a structured SQL Server data warehouse to organize manufacturing, supplier, purchasing, and inventory data for analysis.

Type	Table	Purpose
Dimension	warehouse.dim_product	Product information
Dimension	warehouse.dim_supplier	Supplier information
Fact	warehouse.fact_machine_operations	Machine operations, operating parameters and failures
Fact	warehouse.fact_purchase_orders	Purchase orders, deliveries, quantities and costs
Fact	warehouse.fact_inventory	Inventory levels, closing stock and reorder levels

■ Analytical SQL Views

analytics.vw_supplier_performance

• Supplier fulfillment rate

• Late delivery rate

• Total orders

• Ordered and received quantities

• Purchase value

• Supplier quality and lead time

analytics.vw_inventory_risk

• Inventory days

• Average closing stock

• Minimum stock

• Reorder-risk days

• Reorder-risk percentage

■ Power BI Dashboards

1■■ Executive Overview

High-level overview of manufacturing performance.

• Total Operations

• Total Failures

• Failure Rate

• Average Tool Wear

• Failure Rate by Product Type, Tool Wear, Torque and RPM

• Failure Count by Failure Type

• Torque × Tool Wear Failure Rate Matrix

2■■ Supply Chain & Supplier Analytics

Analyzes supplier performance, purchasing and inventory risk.

• Total Ordered

• Total Orders

• Average Fulfillment Rate

• Average Late Delivery Rate

• Supplier Fulfillment Rate

• Supplier Late Delivery Rate

• Purchase Value by Supplier

• Supplier Quality vs Lead Time

• Reorder Risk by Warehouse

• Average Closing Stock by Warehouse

• Top 10 Products by Reorder Risk

3■■ Manufacturing Root Cause & Operational Analysis

Focuses on operating conditions associated with machine failures.

• Total Operations

• Total Failures

• Failure Rate

• Average Tool Wear

• Failure Rate by RPM, Torque, Tool Wear and Product Type

• Failure Count by Failure Type

• Torque × Tool Wear Failure Matrix

4■■ ■ AI-Powered Failure Analysis

Power BI AI capabilities were used to identify important machine-failure drivers and high-risk combinations.

Key Influencers

• **Low RPM:** 10.40× higher likelihood of machine failure

• **High Torque:** 9.29× higher likelihood of machine failure

• **High Tool Wear:** 2.65× higher likelihood of machine failure

Decomposition Tree

```
Overall Failure Rate: 3.39%
        ↓
High Torque: 13.69%
        ↓
High Wear: 23.95%
        ↓
Product L: 31.44%
        ↓
Low RPM: 35.96%
```

Important: AI findings represent statistical associations within the dataset and should not be interpreted as causal relationships.

5■■ ■ Business Recommendations

01 — Low RPM Monitoring

Finding: Low RPM is associated with a 10.40× higher likelihood of machine failure.

Recommendation: Investigate low-RPM operating conditions and review machine operating procedures for low-speed operations.

02 — High Torque Control

Finding: High Torque is associated with a 9.29× higher likelihood of machine failure.

Recommendation: Monitor high-torque operations more closely and establish torque limits or alerts where appropriate.

03 — Tool Wear Prevention

Finding: High Tool Wear is associated with a 2.65× higher likelihood of machine failure.

Recommendation: Introduce preventive tool inspection or replacement for high-wear conditions before failure risk increases.

04 — Priority Maintenance

Finding: High Torque + High Wear + Product L + Low RPM → 35.96% failure rate.

Recommendation: Prioritize this combination for preventive maintenance and closer monitoring.

■ Key Business Findings

Manufacturing Metric	Result
Total Operations	10,000
Total Failures	339
Failure Rate	3.39%
Heat Dissipation Failures	115
High Torque + High Wear Failure Rate	23.95%
High-risk AI combination	35.96%
Average Fulfillment Rate	90.01%
Average Late Delivery Rate	40.00%
Total Purchase Value	320,233,500
Warehouse Reorder Risk	1.79% – 1.82%

■ Validation & Testing

The Power BI dashboard was validated against SQL Server analytical views and source data.

Metric	SQL	Power BI	Status
Total Operations	10,000	10,000	PASS
Total Failures	339	339	PASS
Heat Dissipation Failures	115	115	PASS
Failure Rate	3.39%	3.39%	PASS
Average Fulfillment Rate	90.01%	90.01%	PASS
Average Late Delivery Rate	40.00%	40.00%	PASS
Total Purchase Value	320,233,500	320,233,500	PASS

Validation Insight: An initial purchase-order-level fulfillment calculation produced 89.02%, while Power BI showed 90.01%. The difference was traced to the aggregation level. The final dashboard uses supplier-level fulfillment rates from analytics.vw_supplier_performance. After validating the analytical view directly, SQL and Power BI both returned 90.01%.

■ Power BI & DAX

• Total Operations

• Total Failures

• Failure Rate %

• Average Tool Wear

• Failure Counts

• Average Fulfillment Rate

• Average Late Delivery Rate

• Reorder Risk

• Inventory metrics

RPM, Torque, Tool Wear, Product Type, Supplier and Warehouse dimensions are used for interactive segmentation and analysis.

■■ Technologies Used

• SQL Server

• T-SQL

• Data Warehousing

• Power BI

• DAX

• Power Query

• Data Modeling

• AI-powered Power BI visuals

• Data Validation

• Business Intelligence

■ Repository Structure

```
Apex-Manufacturing-Analytics/
■
■■■ README.md
■
■■■ SQL/
■   ■■■ 01_Database_Setup/
■   ■■■ 02_Staging/
■   ■■■ 03_Data_Warehouse/
■   ■■■ 04_Analytical_Views/
■   ■■■ 05_Validation/
■
■■■ PowerBI/
■   ■■■ Apex_Manufacturing_Analytics.pbix
■
■■■ Dashboard_Screenshots/
■   ■■■ 01_Executive_Overview.png
```

```
■    ■■■ 02_Supply_Chain_Analytics.png
■    ■■■ 03_Manufacturing_Root_Cause.png
■    ■■■ 04_AI_Failure_Analysis.png
■    ■■■ 05_Business_Recommendations.png
■
■■■ Documentation/
■    ■■■ Project_Documentation.pdf
■
■■■ Data/
    ■■■ README.md
```

■ Business Impact

• Identify high-risk machine operating conditions

• Prioritize preventive maintenance

• Monitor supplier performance

• Identify delivery-risk areas

• Monitor inventory reorder risk

• Understand failure-mode patterns

• Identify high-risk products

• Support data-driven operational decisions

■ Future Improvements

• Predictive machine-failure modeling

• Machine-failure probability scoring

• Automated maintenance alerts

• Supplier risk scoring

• Inventory demand forecasting

• Time-series monitoring

• Automated Power BI refresh

• Machine-learning-based failure prediction

■■■ Skills Demonstrated

```
SQL
 ↓
Data Warehousing
 ↓
Data Modeling
 ↓
Data Analysis
 ↓
DAX
 ↓
Power BI
 ↓
AI Analytics
 ↓
Validation & Testing
 ↓
Business Recommendations
```

This project demonstrates an end-to-end **Data Analyst / Business Intelligence workflow**.

■ Project Status

PROJECT COMPLETED

End-to-End Manufacturing Analytics Solution

SQL Server • Data Warehouse • Power BI • DAX • AI • Business Intelligence